AGROLOOP — MARKET & COMPETITOR ANALYSIS

Course: Design Experience III (AY 2026–27) | **Student:** Veer Nagda | **Mentor:** Prof. Deepali A. Maste | **Vertical:** Entrepreneurship

1. Market Sizing: TAM, SAM and SOM

• **Total Addressable Market (TAM) — ■ 14,800 Crores:** Pan-India organic municipal solid waste processing (34 million MT/yr) and bio-fertilizer / biochar market across Tier-1/2 urban centers and national agricultural belts.
• **Serviceable Addressable Market (SAM) — ■ 1,420 Crores:** Mumbai Metropolitan Region (3,800 MT/day organic scraps) and Pune corridor bulk generators + 450,000 hectares of horticultural farming lands in Konkan and Western Maharashtra.
• **Serviceable Obtainable Market (SOM) — ■ 48.5 Crores:** AgroLoop 3-year target capturing 5 decentralized bio-refinery hubs in MMR, diverting 27,000 MT/yr from 1,200 commercial kitchens & 450 societies, serving 15,000 smallholders.

2. Customer Discovery Survey Synthesis (N=190 Respondents)

Segment	Sample Size	Key Discovery Insights	Willingness-to-Pay (WTP)
Commercial Kitchens	N = 45	84% face BMC SWM compliance notices. Average scrap: 145 kg/day. Current private hauling is irregular.	93% willing to pay ■2,500–■5,000/mo for audited SWM certificates.
Gated CHS Complexes	N = 80	78% attempted on-site composter drums; 82% abandoned within 90 days due to foul odors and flies.	91% willing to pay ■150–■250/flat/mo for off-site EV collection.
Smallholder Farmers	N = 65	92% report soil hardening & low carbon (<0.4%). Average fertilizer spend: ■31,200/acre/year.	86% willing to purchase certified biochar & vermi @ ■10–■24/kg.

3. Competitor Benchmarking Matrix

Dimension	AgroLoop	Hasiru Dala	Phool.co	Soilcraft	BMC Concession
Feedstock	Food Scraps & B2B	Dry & Mixed Waste	Temple Flowers	Crop Stubble	Mixed MSW
Doorstep QC	**YES (BLE IoT <5%)**	Partial (Manual)	N/A	N/A	NO (>40% Plastic)
Biochar (550°C)	**YES (Carbon Sink)**	NO	NO	YES	NO (Windrow)
Multi-Stream	**4 Streams**	1 Stream	2 Streams	1 Stream	1 Stream
AI Advisory	**YES (Gemini+ICAR)**	NO	NO	NO	NO
Gross Margin	**56.5%**	22.0%	38.0%	32.0%	Negative

4. Porter's Five Forces Analysis

• **Bargaining Power of Suppliers (LOW):** Bulk waste generators face strict legal penalties under SWM 2016; they pay AgroLoop to haul waste.

• **Bargaining Power of Buyers (MODERATE-LOW):** Smallholders need affordable soil amendments; synthetic chemicals are 35% more expensive.
• **Threat of New Entrants (MODERATE):** Physical bio-refinery setup, EV logistics, and municipal certifications form high operational moats.
• **Threat of Substitutes (LOW):** Chemical fertilizers face subsidy cuts; on-site residential composters have an 85% abandonment rate.
• **Competitive Rivalry (MODERATE):** Market is vast, fragmented, and dominated by unorganized low-grade waste haulers.